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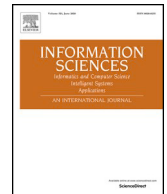


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## FDCNN-AS: Federated deep convolutional neural network Alzheimer detection schemes for different age groups

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Dataset link: <https://www.kaggle.com/datasets/sachinkumar413/alzheimer-mri-dataset>

Dataset link: <https://wwwapp.bumc.bu.edu/fhsresapp/Public/ExportAllToExcel>

Dataset link: <https://github.com/ABDULLAH-RAZA/Alzheimer-Blood-and-Question-Answer-Dataset>

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### ABSTRACT

Alzheimer's disease (AD) is a memory-related disease that occurs in the human brain where neurons become degenerative. It is an evolved form of dementia that deteriorates over time. Machine learning, an extended version of deep learning, has appeared as an optimistic strategy for AD detection. Regardless, the existing AD detection approaches have yet to acquire the expected accuracy, mainly due to unreasonable data for training and testing. In this paper, we present the Federated Deep Convolutional Neural Network Alzheimer Detection Schemes (FDCNN-AS), specifically designed for varying age groups. FDCNN-AS is an efficient framework that contains architecture, algorithm flow, and implementation. It manages AD data from various laboratories and processes it in additional clinics. Our method mixes training data models from different types of data such as positron emission tomography, summed tomography, magnetic resonance imaging, blood tests, and questionnaires about synaptic degeneration. Further, we look at some restrictions that have yet to be addressed in AD detection. These include seeing AD at different ages, extrapolating the severity of brain damage, comparing treatment and recovery rates, and finding benign and malignant ranges in AD data that has been collected. To ensure secure and privacy-preserving learning, we execute FDCNN-AS within a federated learning environment that concerns considerable laboratories and clinics. Within this setup, we operate the generic deep convolutional neural network. The experimental results indicate that FDCNN-AS performs

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## 1. Introduction

These days, in increasing order of ages, the ratio of Alzheimer's disease (AD) has been increasing day by day [1]. AD is a memory- and brain-related disease. It can be a very dangerous disease when humans reach a different age range [2]. AD has different stages such as mild, moderate, and severe. Therefore, to prevent long-term harm, it is very important to detect AD as early as possible [3]. Recently, many digital treatments have helped affected people identify AD in its early stages [4]. Many clinics have used an artificial intelligence approach to detect AD and its stages in human beings [5].

Deep learning has made remarkable progress in recent years when it comes to predicting AD using various datasets. Specific deep learning approaches [6–10] successfully detected AD by analyzing MRI data. Magnetic-resonance imaging (MRI) is a widely used medical imaging technique that combines magnetic fields and adaptive radio to get an image of the entire body infrastructure. It diagnoses the soft tissue and dead cells of the brain in an abnormal form. The CNN and LSTM methods [11–15] are widely exploited in AD with different features based on image data and text in different time predictions. Medical and computer vision-based images gained much popularity with deep learning and convolutional neural networks (CNN). These studies [12–15] investigated the fixed benchmarks of AD with the patient age group between 40 and 85. These studies trained their models on deep convolutional neural networks (DCNN) with different layers. From mild to very mild level prediction with the deep learning methods was achieved in [16–18]. However, AD is an increasing disease among humans, so machine learning methods only work on massive data and are trained according to given data. Therefore, data collection is the biggest challenge for AD. The main reason is that existing methods are only trained and suggested based on similar data types that are publicly available in different repositories. Therefore, the main question is collecting data from various sources without violating data security and privacy during analysis in the designated system. It was suggested in these studies [19,20] that federated learning could help predict AD. It also fixed the security and privacy problems that came up with locally trained data samples using an aggregated node for prediction.

However, there were numerous unresolved research questions regarding AD schemes and the need for more accurate predictions considering the new constraints. (i) Current AD strategies based on deep learning [16–18] primarily focus on utilizing MRI images while neglecting several constraints. The complexity of AD called for solutions that go beyond relying solely on the MRI dataset. These deep learning methods, which heavily relied on the MRI dataset, have largely overlooked important constraints such as AD treatment, recovery ratio, damage area, and differentiation between benign and malignant forms. (ii) In addition, federated learning techniques for AD [19,20] have attempted to combine blood biomarkers with MRI data to achieve more precise predictions. However, these methods predominantly utilized pre-trained MRI datasets with recurrent neural networks (RNNs) and convolutional neural networks (CNNs), resulting in limited data diversity. Consequently, such methods could not be effectively applied across different laboratories with diverse datasets. (iii) Existing AD processing systems [21,22] primarily relied on single-node CNNs. However, collaboration among multiple hospitals was necessary to obtain detailed AD results from different patient data with varying diversity. As a result, the proposed AD system for execution fell short of adequately processing data fusion and multi-modal AD datasets.

This paper introduces the Federated Deep Convolutional Neural Network Alzheimer Detection Schemes (FDCNN-AS) designed for different age groups. We have utilized diverse AD datasets and trained models to achieve optimal accuracy, focusing on specific constraints such as treatment, recovery ratio, damage area, and distinguishing between benign and malignant cases. The contributions of this research can be summarized as follows:

- We introduce the FDCNN-AS, which encompasses architecture, algorithms, and implementation. The architecture illustrates how to input and process data considering the given constraints. It comprises various components, including laboratory, clinical, and remote hospital solutions.
- To address questions (i) and (ii), we propose a federated learning scheme that trains and tests AD datasets with different variables in clinics. The primary advantage is the integration of diverse healthcare clinics and the acquisition of detailed data for AD prediction.
- In response to question (iii), we develop distinct DCNNs on each clinical node to train and test data through local iterations until the desired accuracy for AD prediction is achieved. We have incorporated a security layer into the DCNN to encrypt and decrypt dataset weights before offloading and after downloading data from aggregated nodes. This ensures compliance with the security and privacy requirements of clinic data.

This paper is organized into multiple sections. In Section 2, an in-depth exploration of the related works is presented. Section 3 introduces the AD system proposed in this study, along with its components and an accompanying mathematical model. Section 4 provides a detailed overview of the FDCNN-AS schemes and their solutions, presented in a step-by-step manner. Section 5 focuses on the simulation parameters and implementation details. The result analysis and discussion are covered in Section 6. Finally, Section 7 offers a conclusion that summarizes the obtained results, objectives, and future directions, highlighting the new contributions made in this research.

**Table 1**  
Existing AD Systems and Schemes.

| Study    | Problem      | Method                 | Dataset         | Constraints | Nodes  |
|----------|--------------|------------------------|-----------------|-------------|--------|
| [1–5]    | AD Detection | RNN                    | Biomakers-Blood | Accuracy    | Single |
| [6]      | AD Detection | ANN                    | MRI-epilepsy    | Accuracy    | Single |
| [7]      | AD Detection | DNN-Auto-Encoder       | MRI             | Accuracy    | Single |
| [8]      | AD Detection | CNN                    | Multi-Classes   | Accuracy    | Single |
| [9]      | AD Detection | VGG-TSwinformer        | ADNI            | Accuracy    | Single |
| [10]     | AD Detection | CycleGAN               | MRI             | Accuracy    | Single |
| [11–18]  | AD Detection | CNN-LSTM               | MRI             | Accuracy    | Single |
| [19–22]  | AD Detection | Federated-Learning-CNN | MRI             | Accuracy    | Single |
| [23–38]  | AD Detection | Transfer-Learning-CNN  | MRI-PET         | Accuracy    | Single |
| Proposed | AD Detection | Federated-Learning-CNN | MRI,EEG,Blood   | Many        | Many   |

## 2. Related work

AD is a complex disease, and its exact cause can be predicted to avoid any further brain loss. These studies [1–4] presented single-cell, microbiome, continuous feature, and multi-omics-based solutions to predict AD. These studies demonstrated the deep neural network that extracts the component from MRI images. They determined the level of AD (e.g., non, mild, moderate, and very mild) on the single constraint AD dataset. Furthermore, this study [5] determined AD based on blood omics with different biomarkers by exploiting the Firth Logistic Regression technique. However, the study considered numeric blood values to distinguish AD compared to previous studies. Table 1 provides a summary of the constraints, methods, datasets, research domain, and objectives of the studies. The detailed description of each study, including their methods and objectives, is presented in Table 1. Our proposed work (Table 1), takes into account numerous constraints, incorporates multiple nodes, and employs different techniques compared to existing studies.

The temporal lobe epilepsies for MRI Alzheimer's have been detected in [6]. The study presented a deep neural network-based classifier to analyze the AD data and predict the stages of disease among the 30 to 70-year-old age group. These studies [7,8] presented dense layer classifiers for multi-class MRI datasets and predicted AD among older people. These studies presented the discrete softmax function to decide the runtime on the trained dataset on primary computing devices. The VGG-TSwinformer and VGG16 deep learning approaches are presented in [9,10]. These studies controlled the Relu and Gaussian functions on convolutional layers' and considered only limited MRI image constraints such as mild AD prediction in their models. However, AD has more complex issues, as recently reported in different clinics.

The convolutional neural network and cross-cohort generalizable based on deep learning with the MRI training images enabled techniques presented in [11,12]. These studies divided the features as the gradients were updated with weights and adapted in the classifier at compile and runtime. The goal was to analyze AD at different stages during compilation and runtime. These studies implemented VGG19-enabled strategies to allow the softmax condition to generate the probability of AD at different time-series predictions.

The embedded feature selection and dementia diagnosis from MRI images are aware approaches presented in these studies [13–15]. For these studies, an autoencoder and an ANN with the backpropagation scheme would work well with a deep network that has dense and convolutional layers and activation functions. The objective was to minimize the risk of AD and detect dementia at different levels in different age groups. These studies used deep neural networks, artificial neural networks, autoencoders, recurrent neural networks, long short-term memory, and generative adversarial networks (GAN) to find ways to solve the AD problem. Furthermore, to avoid resource constraints and imbalanced data, these studies suggested federated Alzheimer techniques [19,20], where they only focused on privacy and security. However, they considered the fixed constraints in their models. Therefore it is hard to find the exact cause of AD and its impact on memory and other brain parts in different age groups.

These studies [23–27] successfully integrated transfer learning and CNN to train and evaluate various versions of MRI datasets, including 3D and multiscale formats. One notable advantage of these techniques is their capability to facilitate multimodal data fusion in the analysis of AD, allowing for the extraction of correlated features from the provided data. Transfer learning played a significant role in these studies by enabling the training of models across different nodes. However, it is important to acknowledge that the primary focus of these studies was MRI data and its different feature levels within the context of multimodal data fusion. Additionally, other studies [28–40] proposed a cost-effective approach for AD detection based on hybrid classification using multimodal data. These studies incorporated different datasets, such as MRI and PET, employing 3D classification in their models. The models were trained and tested using distributed layer networks, with the accuracy threshold serving as the given criterion. However, it is worth noting that many of the constraints we are currently discussing have not been fully addressed in these studies.

To the best of our knowledge, the aforementioned studies still do not adequately address several crucial constraints related to AD including treatment, recovery ratio, damage area, and differentiation between benign and malignant cases. The existing AD datasets available are insufficient to encompass these constraints, and the classifiers employed fail to effectively capture the severity of AD disease across different age groups of patients. In light of these gaps, we propose the use of federated deep convolutional neural specifically designed for different age groups. Our objective is to address the aforementioned constraints and accurately detect AD within various patient age groups. The FDCNN-AS schemes are developed with the aim of tackling the challenges of AD detection comprehensively. We emphasize the importance of delving deeper into the detection of AD to mitigate the risk of further memory loss in patients of diverse backgrounds.

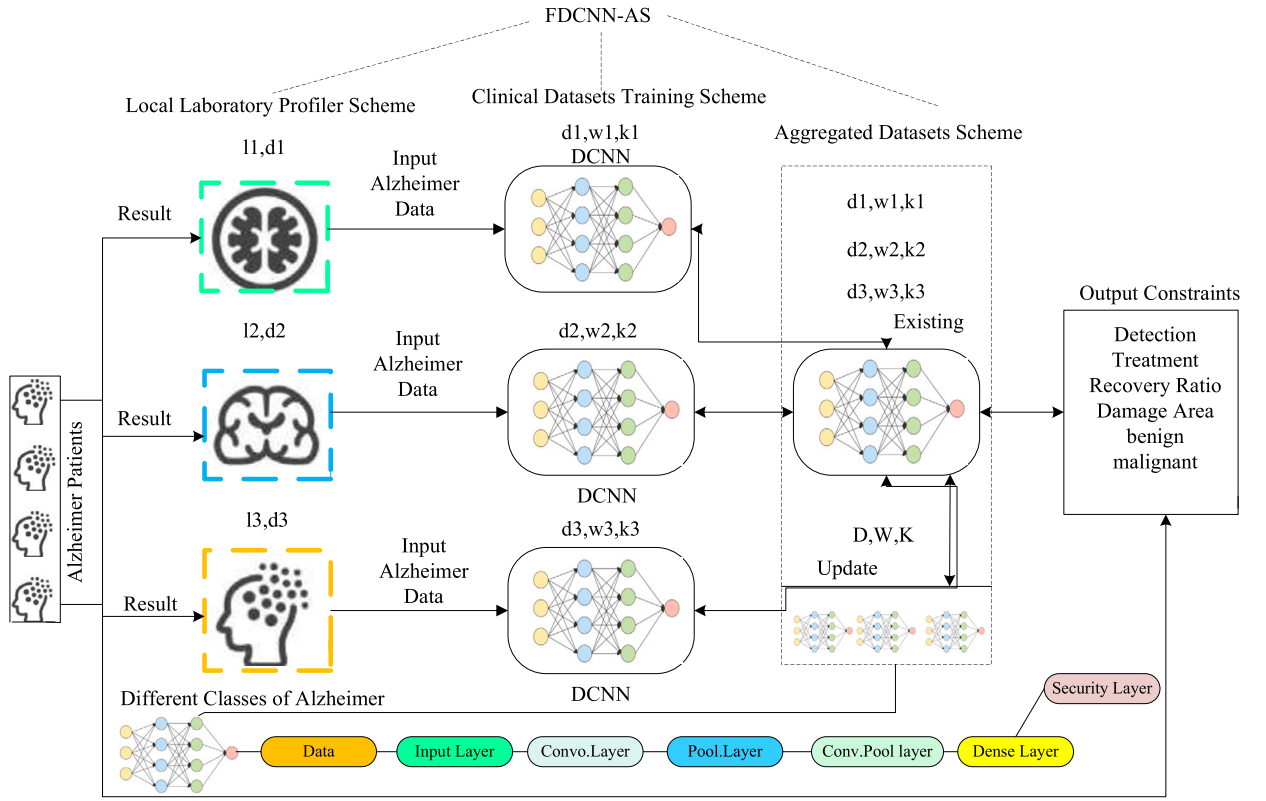


Fig. 1. Proposed Alzheimer Disease Prediction Architecture.

### 3. Proposed Alzheimer disease prediction architecture

The study presents the AD prediction architecture based on different components, as demonstrated in Fig. 1 consisting of different processing from AD patients to clinical decisions. In the FDCNN-AS architecture, three different datasets are integrated and generate real-time data based on AD patients' diagnoses at the laboratory department. We defined the entire process of the FDCNN-AS architecture in the following way.

In the input layer, all the offloaded data will be processed. All the clinics are training their datasets based on new and available data. We have non-linear data. Therefore, the input layer takes the data in different forms such as images and numeric formats. We exploited the convolutional layer to extract the features of the data based on types. The pooling layer obtained the pool of filtered data features and moved them to the next convolutional layer for further feature extraction in the network. The dense layer applied the encryption and decryption mechanisms and generated new weights for the generated data. The security layer ensures the security of all generated weights in the datasets. At the end of each clinic, the softmax function offloads its trained data weights to the aggregated node for further analysis. Table 2 describes the notations of the problem formulation in the mathematical form. The proposed FDCNN-AS algorithm in architecture consisted of different schemes. For instance, Local Laboratory Profiler Scheme, Clinical Datasets Training Scheme and Aggregated Datasets Scheme. Therefore, the AD detection problem is to be solved in different steps.

The data on AD was collected based on different tools at different healthcare laboratories. For instance, computed tomography (CT) [41], magnetic resonance imaging (MRI) [42], positron emission tomography (PET) [43], blood tests [44] and interviews, and synaptic degeneration [45] (Fig. 1). We assumed that architecture considers the  $L$  number of laboratories and dataset  $D$ . Each laboratory  $l$  can process the AD data locally for only collecting and offloading purposes. Each dataset  $d$  has a distinct number of features  $F$ . Therefore, the local processing mechanism of AD patients is determined in the following way.

$$L_d^e = \sum_{p=1}^P \sum_{l=1}^L \sum_{d=1}^D \frac{p \leftarrow d, f, l}{s_l}. \quad (1)$$

Equation (1) shows that the clinics collected the data from patients with features, e.g.,  $p \leftarrow d, f, l$ . Equation (1) generates and collects the data from different patients.

In this paper, we assumed that there are  $K$  number of clinical nodes, where  $k$  is the distinct clinical node with computing capability,  $s_k$ . To train and validate the datasets, we assumed that, at the infrastructure level, we implemented the federated learning on different clinics, for instance, the  $d, w, k$  federated clinical node, where original data offloaded from clinics are trained and validated

**Table 2**  
Problem Mathematical Notations.

| Symbolling                   | Objective                                                        |
|------------------------------|------------------------------------------------------------------|
| $P$                          | Number of Alzheimer Patients                                     |
| $L$                          | Collection of laboratories                                       |
| $l$                          | Distinct laboratory                                              |
| $s_l$                        | Execution speed of laboratory                                    |
| $D, F$                       | Sum of all datasets and Features                                 |
| $d$                          | Particular dataset                                               |
| $T$                          | Total Number of Rounds between local clinics and aggregated node |
| $Z$                          | Local iteration for training datasets                            |
| $\alpha_d$                   | Learning rate of dataset                                         |
| $f$                          | Particular features of dataset $d$                               |
| $R$                          | Collection of Results                                            |
| $r$                          | Particular result of patient                                     |
| $W$                          | Weights of trained datasets                                      |
| $w$                          | Particular weight of dataset                                     |
| $K$                          | Computing Clinical capability                                    |
| $k$                          | Particular clinical hospital                                     |
| $gk$                         | Represents the aggregated node                                   |
| $s_k$                        | Execution speed of clinical node                                 |
| $ag$                         | Aggregated collection of weights                                 |
| $s_{ag}$                     | Speed of aggregated node                                         |
| $Conv$                       | Convolutional layers                                             |
| $Pool$                       | Pool Layers                                                      |
| $conv$                       | Particular convolutional layer                                   |
| $pool$                       | Particular pool layer                                            |
| $FC$                         | Fully connected layers                                           |
| $fc$                         | Particular fully connected layer                                 |
| $\beta, \gamma, ratio, area$ | Benign, malignant, recovery and damage area                      |
| $TP, FP$                     | Detection true and false positives                               |
| Enc-key, Dec-key             | Encryption and decryption keys                                   |

locally. All the local clinically trained datasets must be secured before offloading the aggregated node for further computation. There are two types of iterations included in the architecture. The  $T$  shows the number of iterations between aggregated and local clinical nodes with the updated weights of the datasets. The aggregated node can share all connected nodes. Furthermore,  $Z$  represents the number of iterations during local training and validating data at the local clinics. We integrated the deep convolutional neural network scheme (DCNN) on each clinical node to train and validate the data. The aggregated node is also applied to DCNN for the final decision. The local clinical node determined the processing in the following way.

$$C_d^e = \sum_{k=1}^K \sum_{conv=1}^{Conv} \sum_{pool=1}^{Pool} \sum_{fc=1}^{FC} \sum_{w=1}^W \sum_{z=1}^Z Enc - key \frac{d, w, k}{s_k} \leftarrow Enc, Dec, \quad d = 1, \dots, D. \quad (2)$$

Equation (2) determined the DCNN process to train the offloaded data on different clinics. Each piece of data is encrypted before being offloaded to the aggregated node based on advanced standard encryption (AES) with 256 bits and decrypted by 256 bits in different nodes. The keys are asymmetric and shared by the aggregated node. We determined the security of data on different nodes in the following way.

$$Enc = \sum_{d=1}^D \sum_{f=1}^F \frac{AES(d, f, enc - key)}{s_l}, \quad w = 1, \dots, W. \quad (3)$$

Equation (3) determines the encryption of local gradient data before offloading to the aggregated node.

$$Dec = \sum_{d=1}^D \sum_{f=1}^F \frac{Dec(Enc, Dec - key)}{s_l}, \quad w = 1, \dots, W. \quad (4)$$

Equation (4) determined the decryption process of the encrypted data on different nodes.

$$AG_d^e = \sum_{r=1}^R \sum_{conv=1}^{Conv} \sum_{pool=1}^{Pool} \sum_{fc=1}^{FC} \sum_{w=1}^W \sum_{t=1}^T \frac{D, W, K}{s_{ag}}, \quad p = 1, \dots, P. \quad (5)$$

Equation (5) determines the aggregation process for final decision for all patients.

We formulate the given constraints in the following way.

$$TP = \sum_{d=1}^D AG_d^e w, d, ag_k, \beta, \gamma, ratio, area, Z, \quad p = 1, \dots, P. \quad (6)$$

Equation (6) determined true positive on given input data in the aggregated node.

$$TF = \sum_{d=1}^D AG_d^e w, d, ag_k, \beta, \gamma, ratio, area, Z, \quad p = 1, \dots, P. \quad (7)$$

Equation (7) determined true positive on given input data in the aggregated node. Therefore, detection on aggregated nodes is determined in the following way.

$$Detection = \frac{TP}{TP + Z}, \frac{TF}{TF + Z}. \quad (8)$$

Equation (8) determines the accuracy of AD detection in patients.

#### 4. Proposed FDCNN-AS: algorithm steps

In this paper, we propose the FDCNN-AS algorithm, which consists of different schemes. For instance, Local Laboratory Profiler SchemAlzheimer's data from local clinics into an aggregated node for the e, Clinical Datasets Training Scheme and Aggregated Datasets Scheme. The objective is to solve the AD detection problem with different datasets. We presented the local patient laboratories' data processing with different gathering tools for clinical data gathering and offloading. For the data training and testing on the local clinics, we presented the federated learning deep convolutional neural network scheme with additional functions to support the data privacy and security of the patient's data before sharing it to the aggregated healthcare node for final processing (Algorithm 1).

---

##### Algorithm 1: FDCNN-AS: Algorithm Steps.

---

```

Input :  $D, L, K, W$ 
1 begin
2   Step-1: Local Laboratory Profiler Scheme;
3   Determined local laboratories  $L, D$  based on equation (1);
4   Step-2: Clinical Datasets Training Scheme;
5   Determined clinical processing  $K, D, W$  based on equation (2);
6   Step-3: Aggregated Datasets Scheme;
7   Determined aggregated  $L, K, W, D$  based on equation (5);
8 End Main;

```

---

Three phases are required to process the AD data from local clinics into an aggregated node for a final decision. All the results are returned to the inpatients' laboratories for further analysis based on generated data.

##### 4.1. Local laboratory profiler scheme

In our architecture, the local laboratory scheme collects data on AD from different tools. These AD data are from different laboratories, e.g.,  $L$  that exist in different places. We denoted the generated data as part of different datasets, e.g.,  $D$ . The study presents the local laboratory profiler scheme, which collected laboratory data and offloaded it to the appropriate clinics for further processing. As shown in Algorithm 2, all the patients have access to the different laboratories. For instance, the  $P$  number of users accessing the different laboratories are shown as  $L$ . Algorithm 2 determined the local processing based on equation (3). This expression determined the local processing, e.g.,  $d[p, l, d] = \frac{p-d, l, d}{s_l}$ . Algorithm 2 exploits the profiling, randomly looking to available clinics for offloading the collected data for further processing. For instance,  $clc.status$  is the super global array variable that brings the status of all available clinics based on Algorithm 3 for processing. Furthermore, if the clinics are available and the status becomes 1, e.g.,  $clc.status = 1$ , then the collected data is offloaded to the appropriate clinics for processing.

##### 4.2. Clinical datasets training scheme

In this session, we present the Clinical Dataset Training Scheme based on federated learning. DCNN (FDCNN) is to train and test the Alzheimer Datasets schemes. We trained and tested all AD datasets based on FDCNN in different clinical nodes. Federated learning allows us to train and test different datasets at local clinics and aggregate them on the central node for processing. The different laboratories offloaded data to the appropriate clinics for processing (Fig. 2). We implemented the federated deep neural network on each connected clinic which consisted of different layers (Fig. 2). In the input layer, all the offloaded data will be processed. All the clinics are training their datasets based on new and available data. We have non-linear data. Therefore, the input layer takes the data in different forms such as images and numeric formats. We exploited the convolutional layer to extract the features of the data based on types. The pooling layer obtained the pool of filtered data features and moved them to the next convolutional layer for further feature extraction in the network. The dense layer applied the encryption and decryption mechanisms and generated new weights for the generated data. The security layer ensures the security of all generated weights in the datasets. At the end of each clinic, the softmax function offloads its trained data weights to the aggregated node for further analysis. For the Alzheimer dataset security, we added a new layer in FDCNN, the security layer, which encrypted and decrypted data before sharing it with the aggregated node and verified it by the softmax function in the distinct clinical node. We exploited the asymmetric security ecosystem to encrypt

**Algorithm 2:** Local Laboratory Profiler Scheme.

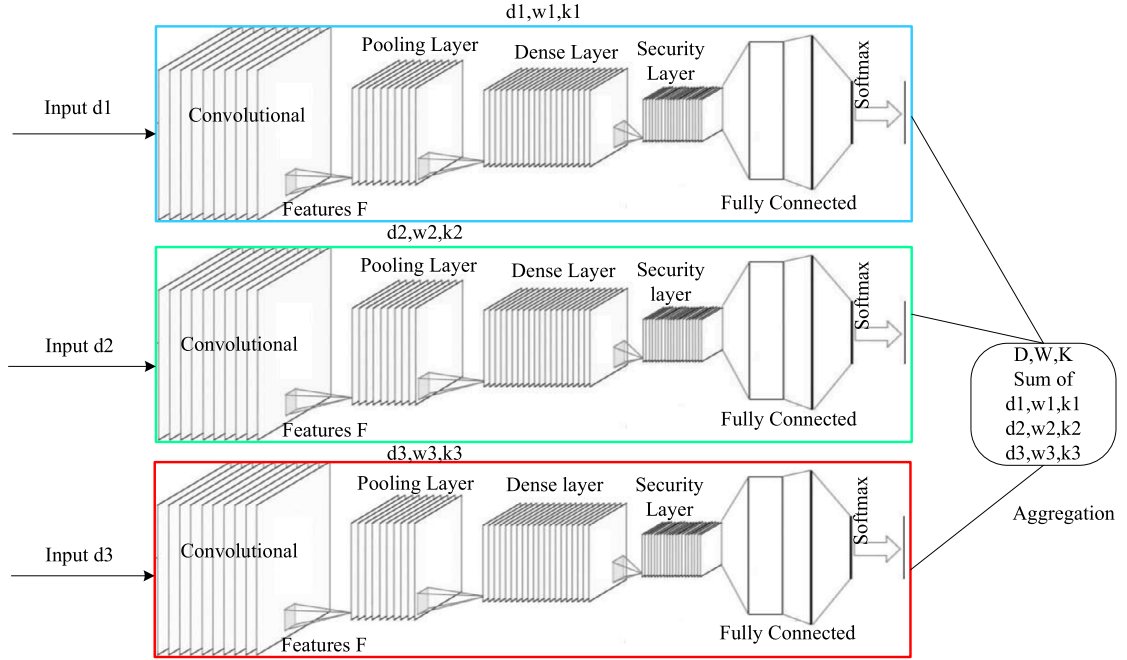
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```

Input :  $D, L, P$ 
1 begin
2   for ( $p, d$  as  $P, D$ ) do
3     Determined the local processing based on equation (1);
4      $d[p, l, d] = \frac{p-d.f.d}{s_l}$ ;
5   Search available laboratories;
6   Call Algorithm 3
7   if ( $clc.status = 1$ ) then
8     Offload  $d[p, l, d]$ ;
9 End Main;

```

---



**Fig. 2.** Clinical Data Processing Training and Testing Based on FDCNN.

trained datasets using advanced standard encryption (AES) [46]. Encrypting the trained healthcare dataset in a distributed clinical environment is not new. Many studies have already done it [46,47]. All the trained and tested weights are encrypted and offloaded to the aggregate for the final decision. It is noted that aggregated data must be capable of quickly decrypting the dataset's data. We integrated the extra layer inside DCNN (the security layer) where all datasets must be encrypted. Softmax validated the data's privacy and security before offloading it to the aggregate for processing (Fig. 2). Aggregated nodes sum all trained dataset weights and make the final decision based on DCNN to predict AD (Fig. 2).

Our algorithm converts different kinds of input data such as images and text into matrixes. Each clinical node trained and tested the appropriate dataset before processing the offloaded data and generating the new weights. We exploited the Tensorflow libraries for training and implemented them in our algorithm. Algorithm 3 trains and tests all the datasets with the new updates on different clinics. We exploited the different hyper-parameters to train and test the AD datasets. For instance, dataset size, layers, features, convolution, pool, and dense layers. Algorithm 3 generated the new weights. However, it is noted that all the weights of newly trained dataset sampling must be encrypted before sharing them with the aggregated node for further processing. We exploited the asymmetric security ecosystem to encrypt trained datasets using advanced standard encryption (AES) [46]. Encrypting the trained healthcare dataset in a distributed clinical environment is not new. Many studies have already done it [46,47]. All the trained and tested weights are encrypted and offloaded to the aggregate for the final decision. It is noted that aggregated data must be capable of quickly decrypting the dataset's data. We integrated the extra layer inside DCNN (the security layer) where all datasets must be encrypted. Softmax validated the data's privacy and security before offloading it to the aggregate for processing. We explain Algorithm 3 in the following way.

- Initially, laboratories generated data offloaded from different sources. All the datasets are split between training and testing, as shown in steps 1 to 6.

**Algorithm 3:** Clinical Datasets Training Scheme.

---

```

Input :  $d[l, d, p] \in D, W, K, \beta, \gamma, ratio, area, Conv, Pool, FC$ 
1 begin
2   foreach ( $d[l, d, p] \in D$ ) do
3     Input layer determined in the following way;
4     Determined the clinical processing time based on equation (2);
5     for ( $z = 1$  to  $Z$ ) do
6       split  $d[l, d, p] \leftarrow k$  into matrixes;
7       Determined the feature extraction in convolutional layer;
8       Determined the encryption and decryption based on equation (3), (4);
9       Trained  $Conv = d[l, d, p, k] \leftarrow F, \beta, \gamma, ratio, area$ ;
10      Test  $Conv = d[l, d, p, k] \leftarrow F, \beta, \gamma, ratio, area$ ;
11      Determined the feature extraction in pooling layer;
12      Trained  $pool = d[l, d, p, k] \leftarrow F, \beta, \gamma, ratio, area$ ;
13      Test  $w.pool = d[l, d, p, k] \leftarrow F, \beta, \gamma, ratio, area$ ;
14      Determined the weights of trained and tested data in the following way;
15       $FC \leftarrow W = w.pool = d[l, d, p, k] \leftarrow F, \beta, \gamma, ratio, area$ ;
16      Probability softmax function for offloading;
17      if  $W = updated$  then
18        All Weights uploaded to the aggregated node;
19    End of Inner Training;
20 End Main;

```

---

- We implemented the DCNN to extract the features, filter them, and process them in different layers such as input, convolutional, pooling, and dense layers, as shown in 7 to 14. The learning rate and other parameters are implemented during the training and testing phases of DCNN.
- Algorithm 3, implemented the security layer, which encrypts and decrypts the data based on AES. We exploited the asymmetric security ecosystem to encrypt trained datasets using advanced standard encryption (AES) [46]. Encrypting the trained healthcare dataset in a distributed clinical environment is not new. Many studies have already done it [46,47].
- All the local datasets are trained and retrained based on different epochs or iterations  $Z$ .
- The softmax function validated that all the trained weights of the dataset are encrypted and ready to offload to the aggregated node.

### 4.3. Aggregated datasets scheme

In this part, we present the aggregated dataset scheme based on federated learning DCNN to aggregate the trained datasets offloaded from local clinics. The goal is to generate the final results based on input and trained datasets for AD detection in networks. Federated learning is the method here in our architecture, which is implemented based on Tensorflow Keras. The goal is to collect clinically locally trained datasets for aggregation and generate the new globally trained model. These healthcare clinical nodes communicate with the federated learning server at least three times per dataset training iteration. The global model of federated learning receives trained weights from clinics. There are two types of iterations (for example, epoch) are included in the architecture. For instance,  $Z$  represents the number of iterations or epochs during the dataset training at the local clinical node until it reaches pre-set accuracy. In the aggregated server,  $T$  is the number of iterations to update the global model based on newly updated data from local clinical architecture. Simply, the aggregated node is the algorithm which performs the average operation on the aggregated data.

**Algorithm 4:** Aggregated Datasets Scheme.

---

```

Input :  $W, DBKey, K$ 
1 begin
2   Aggregated[ $W$ ];
3   for (Every Node Parallel Upload Datasets  $K$ ) do
4     Set Bias for uploaded gradient weights  $B \leftarrow w$ ;
5     foreach ( $w = 1$  to  $W$ ) do
6       Apply the Distributed keys to decrypt all trained datasets;
7        $w \leftarrow DBKey$ ;
8       Determined the aggregated weights based on equation (5);
9        $Agg[w, k, DBkey]$ ;
10      Apply DCNN to extract merged features and decisions based on different layers; To make the decision based on patient input data;
11       $R = D \sim Agg[w, k, DBkey]$  Download generated results from aggregated server;
12    End Aggregation;
13  End Result Sending;
14 End Main;

```

---

**Table 3**  
Simulation Parameters.

| Symbolling          | Objective                                                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P$                 | 15000                                                                                                                                                                   |
| $p$                 | 20 to 60 years                                                                                                                                                          |
| $L$                 | 10                                                                                                                                                                      |
| $l = 1, 2, 3$       | 500 GB Storage                                                                                                                                                          |
| $s_l$               | 16 GB RAM                                                                                                                                                               |
| $D$                 | 3                                                                                                                                                                       |
| $F$                 | 5000                                                                                                                                                                    |
| $d = 1$             | <a href="https://www.synapse.org/#!Synapse:syn2580853/datasets/">https://www.synapse.org/#!Synapse:syn2580853/datasets/</a>                                             |
| $d = 2$             | <a href="https://www.kaggle.com/datasets/sachinkumar413/alzheimer-mri-dataset">https://www.kaggle.com/datasets/sachinkumar413/alzheimer-mri-dataset</a>                 |
| $d = 3$             | <a href="https://wwwapp.bumc.bu.edu/fhsresapp/Public/ExportAllToExcel">https://wwwapp.bumc.bu.edu/fhsresapp/Public/ExportAllToExcel</a>                                 |
| $d = 3$             | <a href="https://github.com/ABDULLAH-RAZA/Alzheimer-Blood-and-Question-Answer-Dataset">https://github.com/ABDULLAH-RAZA/Alzheimer-Blood-and-Question-Answer-Dataset</a> |
| $T, Z$              | 6,000,10,000                                                                                                                                                            |
| $\alpha_d$          | [0.002,0.02, 0.2, 0.1]                                                                                                                                                  |
| $R$                 | 15000                                                                                                                                                                   |
| $W$                 | 38000                                                                                                                                                                   |
| $K$                 | 4, 500 GB, 1000 GB, 3 TB, 10 TB, 16,31,64,128 RAM                                                                                                                       |
| $Conv$              | 64× 64                                                                                                                                                                  |
| $Pool$              | 64× 64                                                                                                                                                                  |
| $FC$                | 128× 128                                                                                                                                                                |
| Activation function | Sigmoid                                                                                                                                                                 |

**Table 4**  
Dataset Training and Features.

| Dataset, Node                  | Training Size | Testing Size | Features | Class | Label Distribution | Encryption     |
|--------------------------------|---------------|--------------|----------|-------|--------------------|----------------|
| $d1, k1$                       | 15000         | 14000        | 2000     | 4     | 1350               | AES-256 bits   |
| $d2, k2$                       | 10000         | 90000        | 2000     | 4     | 1250               | AES-256 bits   |
| $d3, k3$                       | 13000         | 12000        | 2000     | 4     | 1150               | AES-256 bits   |
| $ag_k[k1, d1, k2, d2, k3, d3]$ | 38000         | 35000        | 5000     | 12    | 20000              | AES Decryption |

Algorithm 4 performs the federated learning aggregation process for operation. All the local sampling (called *fusion data*) is aggregated based on decrypted keys in the federated learning from parallel clinical nodes. We applied the DCNN to the merged dataset to get features to make the final decision based on the softmax activation function. The federated learning performed the actions on the patients and sent back the results to the clinic and patients. The  $T$  represents the iterations of updated weights between aggregated nodes and local nodes for data offloading and downloading for execution. These aggregated nodes comprise the DCNN parameters. For instance, these are DCNN and federated learning hyperparameters. For instance, in a local and global iteration of retraining datasets  $T = 6,000, 10,000$ , and learning rate  $\alpha_d = [0.002, 0.02, 0.2, 0.1]$ , patients generated results  $R = 15000$ , and aggregated weighted features and datasets  $W = 38000$ , the total number of clinics  $K =$  are the heterogeneous node that are implemented at different clinics, and convolutional layer  $Conv = 64 \times 64$ , pooling layer  $Pool = 64 \times 64$ , fully connected layer  $FC = 128 \times 128$  and activation function = Sigmoid. The softmax function is the probability and normal distribution where generated, and results are evaluated based on given threshold and bias values.

## 5. Experimental setting and performance evaluation

The study designed the simulation environment based on Tensorflow and Keras and executed it on different high-performance virtual simulators. We configured the simulator configuration file based on different parameters (Table 3). We collected different types of AD data from different patients at different clinics. There are different tools applied at different clinics such as CT scans, MRI, interview sessions, and blood tests, to diagnose AD and the status of memory. The study integrated the different simulation parameters (Table 3). For instance,  $P = 15000$ , age group  $p = 20$  to 60 years, clinics  $L = 10$ , computing capability of clinics  $l = 1, 2, 3 = 500$  GB Storage, and processing speed  $s_l = 16$  GB RAM and datasets and features = 3 and 38000. These are DCNN and federated learning hyperparameters. For instance, local and global iteration of retrain datasets  $T, Z = 6,000, 10,000$ , and learning rate  $\alpha_d = [0.002, 0.02, 0.2, 0.1]$ , patients generated results  $R = 15000$ , and aggregated weights features and datasets  $W = 38000$ , total number of clinics  $K = 4, 500$  GB, 1000 GB, 3 TB, 10 TB, 16,31,64,128 RAM, and convolutional layer  $Conv = 64 \times 64$ , pooling layer  $Pool = 64 \times 64$ , fully connected adaptive layer  $FC = 128 \times 128$  and activation function = Sigmoid. In Table 3, the dataset  $d3$  has two parts. Therefore, we mentioned that  $d3$  has two URLs (Table 3). These studies [11–15] implemented the auto encoder, ANN, and CNN deep learning approaches for AD. For the federated learning training and prediction, we compare our work with the [19,20] studies. We integrated different datasets with different features (Table 4). For instance, dataset, node, training size, testing size, features, class, label distribution, and Encryption. Each dataset is trained and tested at different clinics before being aggregated. For instance, these values,  $d1, k1 = 15000, 14000, 2000, 4, 1350$ , and AES-256 bits, show that dataset 1 will be trained and tested at the node  $k1$  with all the distinct features and values. Furthermore,  $d2, k2, 10000, 90000, 2000, 4$ , and 1250 AES-256 bits show that dataset 2

**Table 5**  
Numerical Results.

| Method       | P     | Age   | Test | Diagnosis | Stage | RR% | DA          | Trm | Accuracy | F1-Score | Recall |
|--------------|-------|-------|------|-----------|-------|-----|-------------|-----|----------|----------|--------|
| ANN          | 10000 | 18-60 | CT   | Mild      | Early | NA  | NA          | 0   | 94.9     | 0.92     | 0.93   |
| Auto-Encoder | 10000 | 18-60 | CT   | Mild      | Early | 75  | NA          | 0   | 94.5     | 0.91     | 0.94   |
| DCNN         | 10000 | 18-60 | CT   | Mild      | Early | NA  | NA          | 0   | 96.4     | 0.963    | 0.966  |
| FDCNN-AS     | 10000 | 18-60 | CT   | Mild      | Early | 81  | Left        | 1   | 98.2     | 0.98     | 0.98   |
| ANN          | 15000 | 21-66 | MRI  | Mild      | Early | NA  | NA          | 0   | 96.2     | 0.93     | 0.94   |
| Auto-Encoder | 15000 | 21-66 | MRI  | Mild      | Early | NA  | NA          | 0   | 94.7     | 0.91     | 0.95   |
| DCNN         | 15000 | 21-66 | MRI  | Mild      | Early | NA  | NA          | 0   | 96.2     | 0.963    | 0.967  |
| FDCNN-AS     | 15000 | 18-60 | MRI  | Mild      | Early | 83  | Left,Right  | 1   | 98.6     | 0.98     | 0.99   |
| ANN          | 13000 | 23-70 | PET  | Mild      | Early | NA  | NA          | 0   | 95.2     | 0.94     | 0.95   |
| Auto-Encoder | 13000 | 23-70 | MRI  | Mild      | Early | NA  | NA          | 0   | 93.7     | 0.92     | 0.96   |
| DCNN         | 13000 | 23-70 | MRI  | Mild      | Early | NA  | NA          | 0   | 97.4     | 0.973    | 0.967  |
| FDCNN-AS     | 13000 | 23-70 | MRI  | Mild      | Early | 83  | Left,Middle | 1   | 99.6     | 0.98     | 0.99   |

**Table 6**  
Numerical Results.

| Method     | P     | $L_e^d$ | $C_d^e$ | $AG_d^e$ | Security | $d1, d2, d3$       | $w1, w2, w3$   |
|------------|-------|---------|---------|----------|----------|--------------------|----------------|
| ANN        | 15000 | 3 hrs   | NA      | 9 hrs    | 0        | 10,000,15000,13000 | 8000,7000,5000 |
| FD [19,20] | 15000 | 3 hrs   | 4 hrs   | 2 hrs    | 1        | 10,000,15000,13000 | 8000,7000,5000 |
| DCNN       | 15000 | hrs     | NA      | 10 hrs   | 0        | 10,000,15000,13000 | 8000,7000,5000 |
| FDCNN-AS   | 15000 | 1hr     | 2 hrs   | 30 mints | 1        | 10,000,15000,13000 | 8000,7000,5000 |

will be trained and tested at the node  $k2$  with all the distinct features and values. Whereas  $d3, k3$ , 13000, 12000, 2000, 4, 1150, and AES-256 bits show that dataset 3 will be trained and tested at the node  $k3$  with all distinct features and values. The aggregated node has the following configuration elements: For instance,  $ag_k[k1, d1, k2, d2, k3, d3]$ , 38000, 35000, 5000, 12, 20000, and AES Decryption show that all datasets are executed on the aggregated node  $ag_k$  with all distinct features and values. We analyzed the numerical results of all methods on different AD datasets. We integrated the different dataset features such as patients  $P$ , age, test, diagnosis, stage, recovery ratio % ((RR), damage area (DA), and treatment recommendation (Trm). We added all the features of datasets during an experiment in the simulation environment. We implemented the different existing machine learning algorithms to analyze the performance of AD detection with different parameters based on different constraints. For instance, the ANN algorithm achieved these results: 10000, 18-60, CT, Mild, Early, NA, NA, 0, 94.9, 0.92, 0.93. Here, the NA means the authors did not consider these parameters for AD prediction in their models. The auto-encoder achieved the following: 10000, 18-60, CT, mild, early, 75, NA, 0, 94.5, 0.91, 0.94. The value of zero means the model considered for AD was not implemented in the previous models. The DCNN is an efficient deep learning algorithm that has optimal results: 10000, 18-60, CT, mild, early, NA, NA, 0, 96.4, 0.963, 0.966. The proposed work is more optimized, generated, and calculated with greater accuracy as compared to the existing method. For instance, 10000, 18-60, CT, Mild, Early, 81, Left, 1, 98.2, 0.98, 0.98. The main reason is that the proposed work considers and implements more constraints in the model as compared to existing studies. The generated and collected AD data obtained optimal results based on FDCNN-AS. Federated learning has the capability to collect heterogeneous feature-related datasets from different clinics in a secure way. The rest of the results for different datasets are shown in Table 5. We determined the processing time at different processing nodes such as laboratories, clinics, and aggregated nodes, as determined in equations (1), (2), (5). For processing the entire dataset, we considered these constraint parameters in the simulation environment. For instance, Method,  $P$ ,  $L_e^d$ ,  $C_d^e$ ,  $AG_d^e$ , Security,  $d1, d2, d3$ , and  $w1, w2, w3$ . We determined the execution process at different computing nodes such as laboratories and clinics, and aggregated it based on the number of hours (Table 6). The results obtained are as follows: ANN, 15000, 3 hrs, NA, 9 hours, 0, 10,000, 15000, 13000, 8000, 7000, 5000 for all datasets of Alzheimer's diseases on different computing nodes. We implemented federated learning as the baseline approach to check the performance of different datasets in the aggregation process due to security constraints. For instance, FD [19,20], 15000, 3 hours, 4 hours, 2 hours, 1, 10,000, 15000, 13000, 8000, 7000, 5000. Federated learning is more optimal during processing than single decision nodes during multiple dataset processing in a distinct clinic. The DCNN obtained the following results: 50005, hours, NA, 10 hours, 0, 10,000, 15000, 13000, 8000, 7000, 5000. The proposed study obtained optimal results as compared to existing studies, as shown in the following. FDCNN-AS = 15000, 1 hour, 2 hours, 30 minutes, 1, 10,000, 15000, 13000, 8000, 7000, 5000.

### 5.1. FDCNN-AS implementation scenario and dataset collection

We designed the simulator environment based on virtualization. We developed the different laboratories' interfaces based on Flask Python [48]. We developed the virtual private server for clinics based on full-stack Python [49]. These provided the sandbox based on the open application programming API and the virtual box for virtualization. We developed the FDCNN-AS based on Tensorflow and Keras and executed the data model on the Google Colab [50] open-source operating system. Thus, we share the datasets and publish sources for further research and analysis. We collected the AD patient data from different healthcare clinics for the research objective

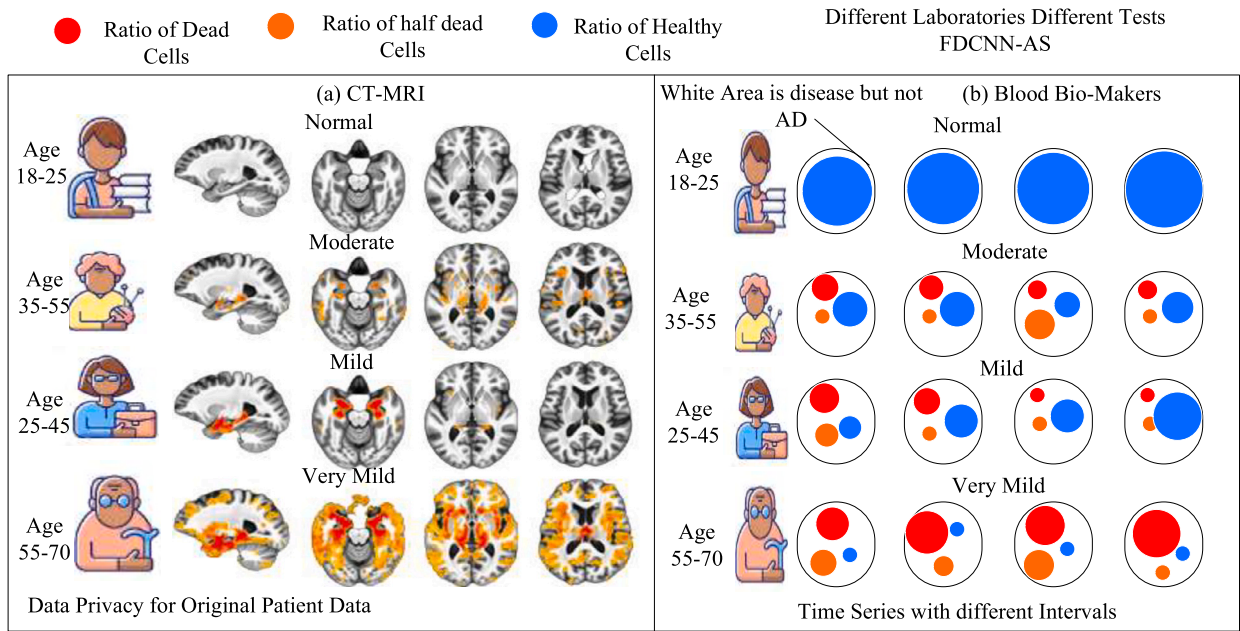


Fig. 3. AD Testing Scenario with FDCNN-AS.

which we uploaded to the following URL: <https://github.com/ABDULLAH-RAZA/Alzheimer-Blood-and-Question-Answer-Dataset>. The rest of the datasets are downloaded from different healthcare repositories (Table 3). We have developed a Federated Learning-enabled environment specifically designed for distributed hospitals, where patients of different backgrounds undergo brain tests for analysis. In this study, we utilize the brain structure concept as defined in a previous publication [38], measuring it in 3D to establish the patterns of AD. However, we have modified the brain structure according to our own architecture which sets us apart from existing AD frameworks and algorithms. Our scenario is optimized to cater to patients of varying age groups (Fig. 3). The proposed FDCNN-AS framework is trained and tested on AD using different combinations of data sources including MRI, CT scans, blood biomarkers, and interviews. To ensure patient security and privacy, we maintain confidentiality regarding the identities of individual patients within the laboratory setting. We conducted visits to hospitals and collected data from diverse age groups ranging from 18 to 70 years old. The MRI and CT scan datasets were categorized into classes such as normal, mild, moderate, and very mild, enabling us to identify the presence of AD in specific areas of the brain and comprehend its structural characteristics in various forms. These images were captured in 3D format allowing us to analyze and identify patterns of AD in the brains of individuals belonging to different age groups. Additionally, we collected blood test results from multiple laboratories, organizing the data into different range cells. The blue cells represent a healthy state, the orange cells indicate a partially deceased ratio of cells, and the red cells signify a fully deceased ratio. Furthermore, white spaces within the brain indicate cell deficiencies elsewhere in the body, potentially indicating the presence of other diseases that may manifest as headaches.

As per Fig. 3 (a) and Fig. 3 (b), patients of various age groups visit the laboratories at different intervals, undergoing MRI, CT scans, and blood biomarker tests at different time points. We have observed that the incidence of AD varies across different age groups. For instance, individuals between the ages of 18 and 25 undergo different tests in distinct dimensions and at varying intervals. Our analysis indicates that they may be experiencing headaches and temporary memory loss due to another underlying condition. Upon examining the blood cells, we have noticed that all cells appear to be healthy, but the presence of white space suggests some deficiency. Consequently, it becomes clear that a single test is insufficient for determining and detecting AD. Patients who exhibit temporary memory loss and display deficiency in cells may present symptoms similar to those of AD. However, it is important to note that they do not have AD, and the recommended medications and treatments may not be suitable or accurate for their condition. After analyzing different age groups, it becomes apparent that individuals between the ages of 35 and 70 have a higher susceptibility to developing AD, with variations in cell patterns and different disease classes. It is important to note that once cells are deceased, they cannot be revived, and partially damaged cells cannot be fully restored. However, it is possible to prevent further cell loss and maintain the functionality of healthy cells (represented by blue cells) within the patients' brains. Furthermore, it is worth noting that patients experiencing white space and areas in their brains, along with headaches and temporary memory loss due to deficiencies, face a unique challenge. In these cases, medical care providers may mistakenly recommend the same medications prescribed for AD patients which can have adverse effects on the health of these individuals.

## 5.2. Description of datasets

AD is a complex disease, and different datasets are collected from healthcare institutions based on real patients of all genders with the specified specifications. In the simulation, we gave the URLs of all datasets. All collected datasets' URLs are given in Table 3.

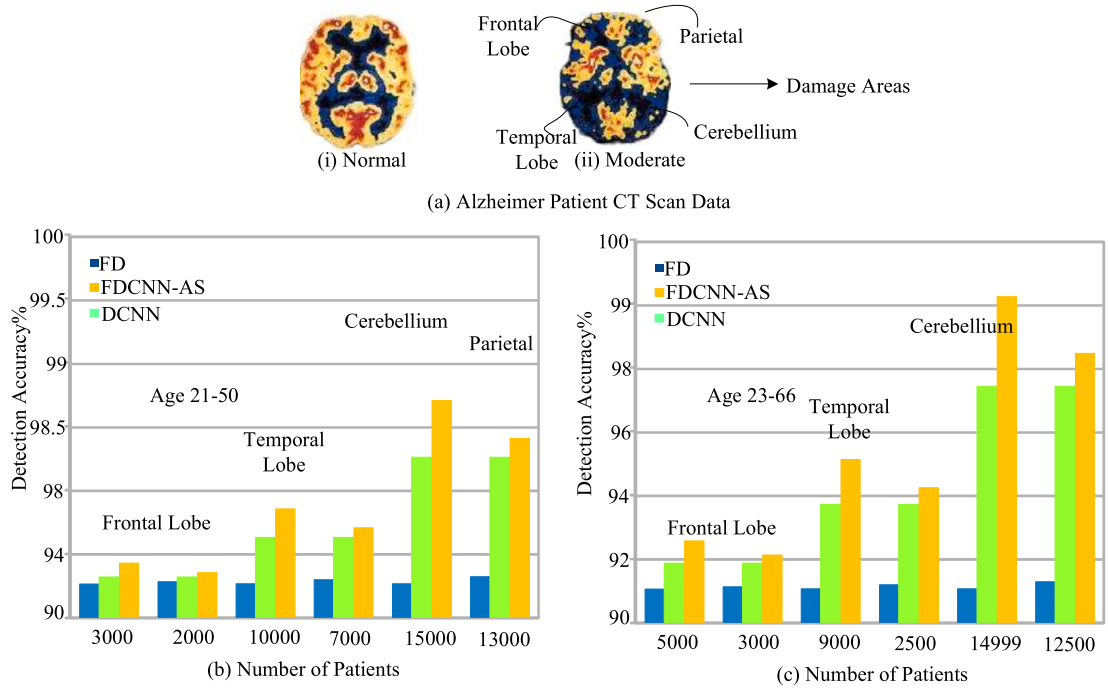


Fig. 4. Alzheimer Detection with Different Brain Parts and Accuracy Obtained by Different Methods.

We integrated different datasets with their different features (Table 4). We collected the AD patients' data from different healthcare clinics for the research objective, which we uploaded at the following URL: <https://github.com/ABDULLAH-RAZA/Alzheimer-Blood-and-Question-Answer-Dataset>. The rest of the datasets are downloaded from different healthcare repositories, and their URLs are given in Table 3. The datasets consist of different images, numeric values, and text data collected at varying laboratories from different patients. Furthermore,  $d2, k2, 10000, 90000, 2000, 4$ , and 1250 AES-256 bits show that dataset 2 will be trained and tested at the node  $k2$  with all the distinct features and values. Whereas  $d3, k3, 13000, 12000, 2000, 4, 1150$ , and AES-256 bits show that dataset 3 will be trained and tested at the node  $k3$  with all distinct features and values. The aggregated node has the following configuration elements. For instance,  $ag_k[k1, d1, k2, d2, k3, d3]$ , 38000, 35000, 5000, 12, 20000, and AES decryption show that all datasets are executed on the aggregated node  $ag_k$  with all distinct features and values. We integrated the blood tests and question-and-answer suggestion dataset with the  $d3$  as available at <https://github.com/ABDULLAH-RAZA/Alzheimer-Blood-and-Question-Answer-Dataset>.

## 6. Results and discussion

AD is a complex disease. Therefore we collected the different datasets, which were generated with different tools. In the simulation, we evaluated the performance of the models based on different data. For instance, we take the CT scan AD image of random patients (Fig. 4 (a)). The image has two parts such as Fig. 4 (a) (i) shows the normal image and Fig. 4 (a) (ii) shows an abnormal CT scan AD image. During the analysis and processing, we detected AD in different parts of the brain. For instance, Frontal Lobe Temporal, Lobe Cerebellum, and Parietal Frontal Lobe and determined the damaged areas of the brain due to different AD features (Fig. 4 (a) (ii)). We implemented deep learning and federated learning as the baseline approaches. For instance, federated learning (FD), FDCNN-AS, and DCNN. We analyzed and experimented on different ranges of patients such as 3000, 2000, 10000, 7000, 15000, and 13000, as shown in the x-axis. The y-axis shows the accuracy range of methods: 90, 94, 98, 98.5, 99, 99.5 and 100. In the simulation results (Fig. 4 (b)), we detected all parts of AD in patients between the ages of 21 and 50. We randomly took all gender data to the simulator for evaluation. We gave the 13000 inputs of data, from which we obtained 3000 patients suffering from frontal lobe AD, with an accuracy of about 94. In the second iteration, we obtained the results for the 2000 patients with an accuracy of about 93. We determined the temporal lobe of AD with a random number of patients such as 10000 and 7000, with an accuracy of 96 and 98. Additionally, the number of patients with parietal AD and cerebellum AD varies between 15000 and 13000. However, we analyzed middle-aged people's experiences with the different ADs in the brain. Fig. 4 (c) shows the AD of different patients with the age range of 23 to 66 with different accuracies. It can be observed that FDCNN-AS performed better than FD and DCNN. The main reason is that all existing models considered only four mild classes of AD with the same type of data. Whenever these methods are tested with heterogeneous clinical nonlinear data, they lose accuracy as compared to FDCNN-AS. We implemented the different deep learning techniques to determine the AD class among different random patients (Fig. 5). There are four techniques that were implemented in the simulation environment. For instance, FDCNN-AS, ANN, CNN and A-Encoder. As per Fig. 5 (a), we determined that all deep learning and federated learning have the capability to determine AD with a different range of accuracy. For instance,

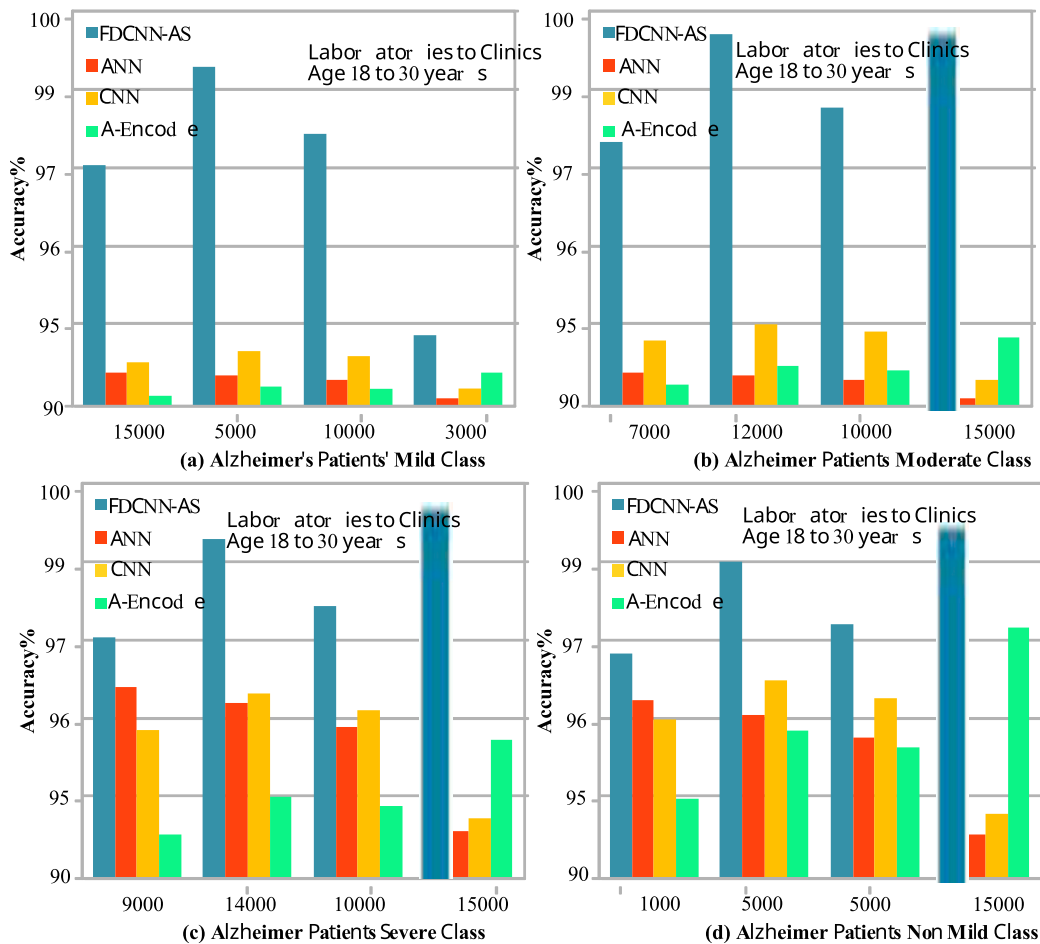


Fig. 5. Alzheimer Disease Detection Accuracy Performance with Different Classes Based on Different Approaches.

Fig. 5 (a) determines Alzheimer's Patients' Mild Class, which means many patients have an early level of AD. We determined the accuracy of 95, 96, 97, 99, and 100. We found that, in different age groups, AD exists in different patients. The proposed algorithm framework, FDCNN-AS, obtained 98% accuracy for 15000 patients when they were identified as having mild class-related AD. The rest of the algorithms achieved 94% accuracy for patients 15000. With the 5000 patients' data, FDCNN-AS obtained 99% accuracy as compared to existing studies, which obtained 95% accuracy during the execution. Fig. 5 (b) (c) (d) shows that young patients also have AD and are thus evaluated during the tests and analyses via different methods. These days, AD affects the overall performance of patients in different age groups. The proposed FDCNN-AS offered a research environment in which the dataset size became huge and helped analyze and detect AD compared to existing methods and datasets.

Fig. 6 shows the benign and malignant forms of AD among different patients. We analyzed the benign and malignant cells in different classes (Fig. 6 (a)), from normal AD, moderate AD, and severe AD (for example, very mild). It is noted that all deep learning models can identify the level of AD in patients. During the experiment, we found that 5000 patients had different memory problems. For instance, as shown in Fig. 6 (b), 5000 patients were afflicted with long-term memory loss, along with a moderate class of AD. However, all methods obtained an accuracy of nearly 93%. At the same time, short-term memory was identified in 5000 patients with 94% accuracy. Random memory loss means the patient may have many issues such as language problems, short-term memory loss, memories of things, and answering questions. We identified these issues as in Fig. 6 (b) with an accuracy of nearly 98%. However, Fig. 6 (b) showed that patients' disease can be treated and controlled at some level, as compared to Fig. 6 (c) where the level of AD and memory loss can not be controlled.

## 7. Conclusion and future work

This paper has presented the AD detection scheme based on different data modalities such as CT scan, MRI, blood biomarkers, and question-and-answer data. The research questions and contributions of the study were presented, highlighting the efforts and limitations of existing studies in addressing the AD problem. The FDCNN-AS, which combined federated learning and deep CNN in fog and cloud networks, was introduced. To demonstrate the effectiveness and efficiency of FDCNN-AS, experiments were conducted

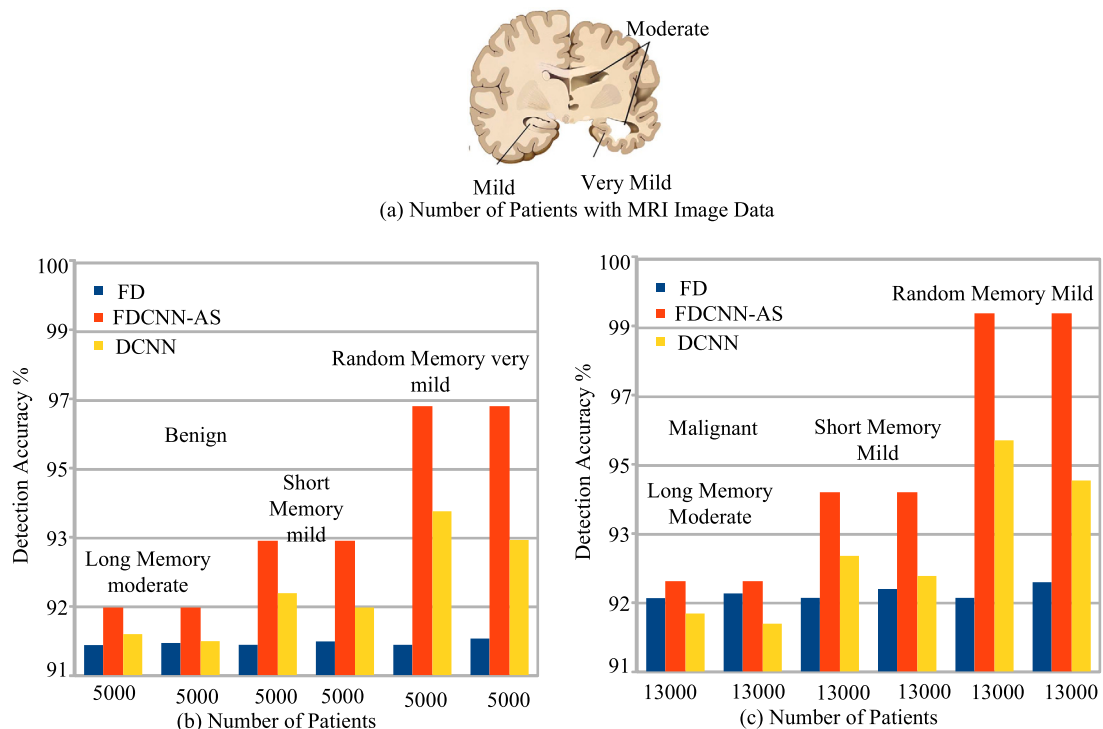


Fig. 6. Performance of Deep Learning Method To Predict Benign and Malignant of Alzheimer Disease Among Patients.

using various data modalities. The experimental and result discussion sections detailed the implementation of different datasets and the relevant parameters. The results and discussion confirmed that all constraints were successfully addressed on different computing nodes, surpassing existing studies. Furthermore, the scenario in which to conduct AD tests with different age groups was discussed, with emphasis on the varying results obtained from different datasets in different time series and intervals. These findings underscore the potential of the FDCNN-AS model to revolutionizing the field of AD detection and treatment.

However, there will be significant improvements in the future version of FDCNN-AS. (i) The future work will expand the scope of federated learning to incorporate a wider range of hospitals with diverse environments, overcoming the current limitation of solely focusing on local clinical settings. (ii) To tackle the heterogeneity among hospitals, a blockchain technology-based solution will be developed for FDCNN-AS, ensuring seamless interoperability and secure data sharing between different nodes. (iii) Moreover, in upcoming research, we will consider the drug recommendation constraints for both AD and non-AD patients, which were not addressed in this study.

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### CRediT authorship contribution statement

**Abdullah Lakhan:** Conceptualization, Methodology, Software, Validation, Writing – original draft. **Mazin Abed Mohammed:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis. **Mohd Khanapi Abd Ghani:** Writing – original draft. **Karrar Hameed Abdulkareem:** Writing – review & editing. **Haydar Abdulameer Marhoon:** Supervision, Writing – review & editing. **Jan Nedoma:** Writing – review & editing, Project administration, Funding acquisition. **Radek Martinek:** Writing – review & editing, Project administration, Funding acquisition. **Muhammet Deveci:** Supervision, Visualization, Writing – review & editing.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

All AD datasets are publically available at the following URL:

<https://www.synapse.org/#!Synapse:syn2580853/datasets/>

<https://www.kaggle.com/datasets/sachinkumar413/alzheimer-mri-dataset>

<https://wwwapp.bumc.bu.edu/fhsresapp/Public/ExportAllToExcel>

<https://github.com/ABDULLAH-RAZA/Alzheimer-Blood-and-Question-Answer-Dataset>

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